

Computing CMB temperature fluctuations for spherical spaces

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Manifolds & Homotopy groups

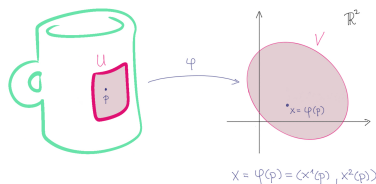


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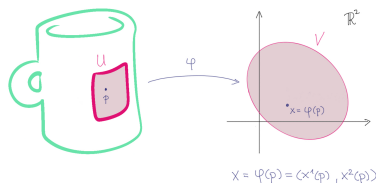


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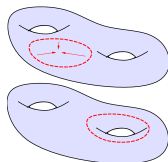


Figure: Two double tori with (non)-contractible paths [7].

Quotients of the 3-sphere

- $SO(4)$ is isomorphic to the isometry group of $\mathbb{S}^3$.
- Subgroups $\Gamma \leq SO(4)$ define equivalence classes of orbits on $\mathbb{S}^3$ by the standard action of $SO(4)$ on $\mathbb{R}^4$, i.e. $x \sim y$ iff $x = My$ for some $M \in \Gamma$.

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- The obtain space is (*sometimes*) a manifold. In particular, it is well defined and spherical for finite Γ .
- This can be easily generalised to the n -sphere.

Lens spaces

- Lens spaces are obtained by taking the quotient of some n -sphere by a cyclic group.
- They cannot be distinguished by their homotopy group alone.

Lens spaces — the explicit construction

Definition (Lens space)

Given $q \in \mathbb{Z}$ and $s \in \mathbb{Z}^n$ elementwise coprime with q

$$L(q, s) := \mathbb{S}^{2n-1} / \langle M_{q,s} \rangle,$$

where $\langle M_{q,s} \rangle$ is the group generated by

$$M_{q,s} := \begin{pmatrix} R(2\pi s_1/q) & & & \\ & R(2\pi s_2/q) & & \\ & & \ddots & \\ & & & R(2\pi s_n/q) \end{pmatrix}$$

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In the $\mathbb{S}^3$ case, we denote $\mathbf{L}(p, q) := \mathbf{L}(p, (1, q))$ for coprime $p, q \in \mathbb{Z}$.

The Laplace-Beltrami operator

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- 2 The spectrum determines many things about the space (like its volume).
- 3 When two distinct (up to isomorphism) manifolds share a spectrum, they form an *isospectral pair*.

The silver lining

- Isospectral pairs are cool and all, but they cannot occur in dimension 3.
- We can thus attempt to infer the shape of our universe based on its spectrum.

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- The set of all possible wave numbers [for which a nonzero solution exists] depends on Γ - usually a proper subset of $\mathbb{N}$.
- Solutions to the Helmholtz equation are calculated for each Γ and CMB anisotropy $\frac{\delta T}{T}$ computed as a sum of spherical harmonics.

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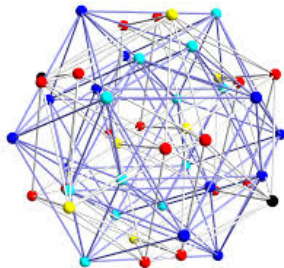


Figure: Graphical representation of the binary tetrahedral group [5]

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 - ① homogeneous: $N3$ and $L(8, 1)$.
 - ② inhomogeneous: $N2 \equiv L(8, 3)$.
- Results: inhomogeneous spaces have more variety in the CMB anisotropies than homogeneous spaces, because the strength of anisotropy suppression is observer dependent.

Statistical Isotropy and Hypothesis

Theoretical Expectation:

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Key Question: - Does the observed CMB data deviate from statistical isotropy? - If $\langle a_{\ell m} \rangle \neq 0$, this suggests a preferred cosmic direction.

Methodology: Sky Masking and the Test Statistic

Challenges in Real Observations:

- We cannot observe the full CMB sky due to foreground contamination.
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Monte Carlo Simulations:

- We generate thousands of **simulated CMB skies** assuming isotropy.
- Each sky has different random $a_{\ell m}$, drawn from:

$$a_{\ell m} \sim \mathcal{N}(0, C_\ell)$$

- The observed WMAP data is compared against these simulations.

Results and Interpretation

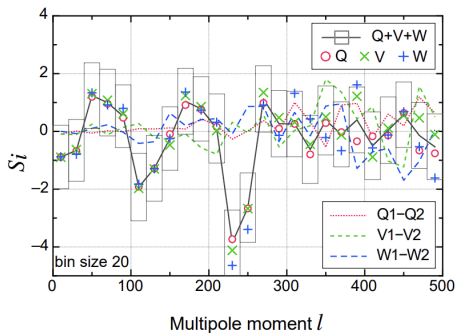
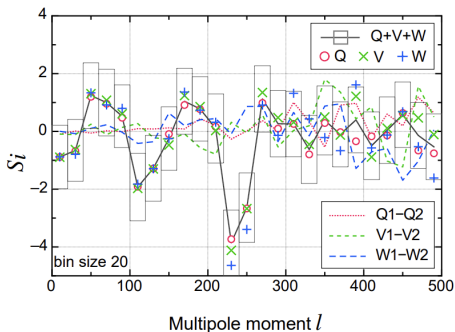


Figure: The decorrelated band mean test statistic values over multipole ranges [9].

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Key Findings:

- A statistically significant deviation was found in $221 \leq \ell \leq 240$.
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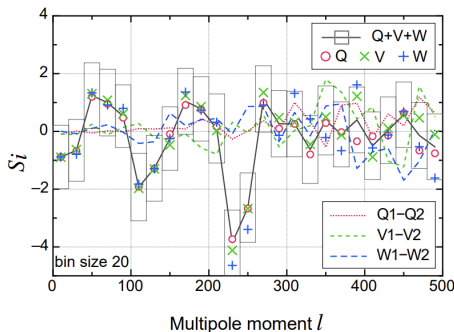


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Possible Explanations:

- A real cosmological signal? → A finite universe or new physics.
- A systematic effect? → Foreground contamination or instrumental noise.

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- ② There are two homogeneous spherical manifolds obtained as $\mathbb{S}^3/\Gamma$ which produce CMB similar to observations.
- ③ Inhomogeneous spherical spaces exhibit varied behavior of CMB anisotropies.
- ④ Statistical test results suggest possibilities of finite multi-connected topology.

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Thank You!